

refuses to answer or doesn't know how to answer. Missing responses in the ESS dataset are assigned codes that are much higher than the possible range of real responses. For example, on a question that asks a respondent to choose a number on a scale from 1 to 10, the ESS records a 77 response if the respondent refuses to answer. For our analysis, we'll consider only responses that are complete, with no missing values for variables that interest us. We can restrict the `ess` data so that it contains only full responses for the variables we care about as follows:

```
ess = ess.loc[ess['sclmeet'] <= 10, :].copy()
ess = ess.loc[ess['rlgdgr'] <= 10, :].copy()
ess = ess.loc[ess['hhmb'] <= 50, :].copy()
ess = ess.loc[ess['netusoft'] <= 5, :].copy()
ess = ess.loc[ess['agea'] <= 200, :].copy()
ess = ess.loc[ess['health'] <= 5, :].copy()
ess = ess.loc[ess['happy'] <= 10, :].copy()
ess = ess.loc[ess['eduyrs'] <= 100, :].copy().reset_index(drop=True)
```

Splitting Our Data

There are many ways we could use this data to explore the relationship between someone's social life and their happiness. One of the simplest approaches is a `binary split`: we compare the happiness levels of people with highly active social lives to those of people with less active social lives ([Listing 9-1](#)).

```
import numpy as np
social = list(ess.loc[:, 'sclmeet'])
happy = list(ess.loc[:, 'happy'])
low_social_happiness = [hap for soc, hap in zip(social, happy) if soc <= 5]
high_social_happiness = [hap for soc, hap in zip(social, happy) if soc > 5]

meanlower = np.mean(low_social_happiness)
meanhigher = np.mean(high_social_happiness)
```

[Listing 9-1](#): Calculating the mean happiness levels of people with inactive and active social lives

In [Listing 9-1](#), we imported the `numpy` module in order to calculate means. We defined two new variables, `social` and `happy`, by slicing them from the `ess` dataframe. Then, we used list comprehensions to find the happiness levels